

Classical Modelling Notes

Levels 1 and 2 for the OCEANS 2026 USV Identification Paper

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Internal working notes. Not for circulation.

This document holds the detail: the annotated reference checklist, the development of the sufficiency argument and the derivation behind the margin-based diagnostic. It is the scratch space for the traditional engineering models, Levels 1 and 2.

The companion `paper.tex` is the team-facing document. It carries the paper outline at outline-level specificity, plus section status and action assignments. Detail belongs here; coordination belongs there. When something here settles, promote a one-line summary into `paper.tex` rather than copying the argument across.

Blue superscript dots mark citations with a local PDF in `refs/`.

Contents

1	Introduction	2
2	Closely Related Work	5
2.1	Response Models as the Minimal Control-Oriented Description	5
2.2	Physics-Based Maneuvering Models and the Identifiability Barrier	6
2.3	Control-Oriented Identification of Small Uncrewed Craft	7
2.4	Data-Driven and Grey-Box Alternatives	9
2.5	Positioning of This Work	10
3	Model Spectrum Methodology	10
3.1	Level 1: First-Order Linear SISO Models	10
3.2	Level 2: Nonlinear State-Space Model	11
3.3	Level 3: Black-Box NARX Model	11
3.4	Cross-Model Diagnostics	11
4	Experimental Platform and Data Collection	12
5	Results	13
5.1	Level 1: First-Order SISO	13
5.2	Level 2: Nonlinear State-Space	13
5.3	Level 3: NARX	13
5.4	Cross-Model Diagnostic Summary	13
6	Discussion	13

1 Introduction

- **Motivate.** Small USVs (under 7 m, single thruster plus rudder) are increasingly deployed for survey, inspection and autonomy research. Low-level speed and heading control is a prerequisite for any higher-level autonomy.
- **The tension.** Control design requires a model, but the “right” model depends on what the model is for. “How much fidelity is enough?” has no answer until “enough for what?” is specified.
 - Because that specification is usually left implicit, the default is a bias toward more fidelity: the benefit of an added term is easy to measure (ΔR^2), while its cost is diffuse and rarely accounted for, namely identifiability from feasible experiments, experimental burden and loss of design insight.
 - **Broadly, this is the overfitting problem.** Driving R^2 up is a naive objective once it is separated from what the model is for. Past some point the added terms improve the fit and damage the thing we actually want, which is good control. Machine learning at least made that trade visible by convention, through held-out data, and pays it down through regularization [1]•.
 - Held-out scoring is a weaker check here than it is in machine learning, because the *parameters* and not only the predictions are the deliverable. A maneuvering model can pass a prediction test while its individual coefficients are meaningless [2]•. Hence diagnostics that compare *across* model structures rather than scoring one structure on held-out data.
 - Prior work is *not* naive about this, and the most careful example shows exactly where the difficulty sits. Sonnenburg and Woolsey [3]• build genuine quantitative machinery for sufficiency: a cost function evaluated on held-out validation maneuvers, tabulated per model and per regime. The final decision is nonetheless made by judgment, discarding the model that scores best because its two extra parameters are “not worth the slight edge” in turn-rate estimation.
 - **This is not a lapse on their part.** Their cost function scores the *benefit* of added parameters. No comparable metric exists for the *cost*, so there is nothing for the judgment to be replaced by. The quantitative apparatus stops one step short of the decision it was built to support. That vacancy, not any author’s rigor, is what this paper is aimed at. *Detail in Sec. 2.3.*
 - The residual gap is that sufficiency is resolved by *selecting* one model, after which the discarded models stop contributing.
- **Framing.** George Box - all models are wrong but some are useful [4]: a spectrum of models, each wrong in different ways, is collectively more useful than any one complex model.
- **Sufficient for what? The decisions a control-oriented model must support.** Models (and simulations) are means to an end, not ends in themselves. The end here is specific: transferring low-level stabilization control from simulation to the vehicle.
 - **The naive premise, stated so we can reject it.** The implicit default is often to assume that the model is faithful replica of the system under test, a twin (analog or digital), so that with sufficient fidelity one could engineer control and autonomy entirely in simulation and

transfer zero-shot to hardware. Fidelity is then an unbounded good and the only question is how much of it you can afford.

- **Fidelity is not the objective, and pursuing it past a point does concrete harm.** What we want is control that works on the vehicle, and that is not monotone in model fidelity. Four costs, none hypothetical:
 - *The estimates get worse, not merely more numerous.* Added terms degrade the conditioning of the fit, so parameters that were previously well determined become less certain [1][•]. That is a loss taken, not a gain forgone.
 - *Field time is the scarce resource.* Every added term needs excitation to identify it, so fidelity converts directly into trials on a platform where trials are the binding constraint.
 - *Design insight degrades.* A model with many terms predicts without explaining. It cannot say which term drives the behavior, which is exactly what architecture selection (D1) needs from it.
 - *False confidence.* A high-fidelity model invites the belief that the computed gains will transfer, which is D5 below and does not hold. Higher fidelity can therefore make a team *less* likely to plan field validation. That is a schedule and safety risk, not a modeling one.
- **Two independent bounds on useful fidelity.** The economic bound below is the obvious one. The sharper claim is that even if fidelity were free there would still be a ceiling: once the model discrepancy falls inside the robustness margin of the design, further fidelity changes no decision and is inert. The limit is structural, not budgetary. *This is the quantity Diagnostic 4 (Sec. 3.4) computes.*
- **Two distinct claims, and the second is the one we need.**
 - *Box’s claim is epistemic.* Correctness is unattainable, so a wrong model cannot be made “‘correct’ . . . by excessive elaboration,” and “overelaboration and overparameterization is often the mark of mediocrity” [5][•]. Parsimony follows from the impossibility of truth.
 - *Right versus effective is a decision claim, and stronger.* Additional correctness is often genuinely available and should still be declined when it trades against the outcome. Skelton put this in control terms in 1989 [6]: model errors “need not be ‘small’ but simply ‘appropriate’ ” for the control design. Fidelity is not the criterion, fitness for the design decision is.
 - Diagnostic 4 sits squarely on the second claim. Level 2 genuinely is a better description of the physics than Level 1, and past the margin threshold it still changes no design decision. Box does not predict that. Skelton does, and neither of them says how to check it on a given vehicle.
- **The gap we address.** Skelton’s criterion has been available for thirty-five years and remains qualitative, since “appropriate” is not a computable predicate. The identification-for-control tradition [7, 8] formalized the idea for experiment design without producing a field test a small-boat practitioner can run. Diagnostic 4 is an attempt at that test for this vehicle class.
- **What sets how much model you need is the economics, not the physics:** the cost of the platform, access to the test environment and above all the ratio of hardware-test cost to simulation-test cost. A cheap vehicle on an accessible lake and an expensive vehicle with a narrow at-sea window warrant different answers to an identical modeling question.
- **What a model of this class *can* support,** in rough order of increasing demand:
 - **D1: choose a control architecture.** Weakest requirement. The model need only get the

dominant structure right: how many states, which couplings matter, where the dominant timescale sits.

- **D2: verify the architecture end-to-end.** Does the control system function, and function correctly, when wired up whole? Catches structural and integration faults on the bench rather than on the water.
- **D3: compare architectures relatively.** What is the cost and benefit of adding a feedforward term or a rate filter? Differences between architectures survive model error far better than absolute values do. Caveat: each candidate must be tuned before it can be compared, so tuning quality confounds the comparison unless explicitly controlled for.
- **D4: rehearse the tuning process.** Practice the procedure and build the tooling that will be used to tune in the field. The deliverable is the tuning workflow, not the gains it produces.
- **What the classical side contributes back.** Gain and phase margin state how much model error a design tolerates. That is a cost-side metric of exactly the kind missing from the sufficiency literature above: once the discrepancy between two model levels falls inside the margin of the controller designed from the simpler one, the extra fidelity buys nothing for that design. The margin is the transfer budget. *Developed as Diagnostic 4, Sec. 3.4.*
- **What it cannot support (D5):** identifying parameters and designing the controller in simulation and then expecting zero- or one-shot transfer. Saying this plainly is unusual, and it is the honest answer to “why model at all if you re-tune in the field anyway.” You are buying D1–D4. You are not buying the gains.
- Our three levels map onto these decisions rather than onto a fidelity ladder: Level 1 serves D1, Level 2 serves D2 and D3, Level 3 bounds how far D4 could be pushed with more data. *Returned to in Sec. 6.*
- **Open.** This ladder merges the earlier purpose framing (choose / compare / parameterize) with the sim-to-real framing. D4 is the piece with no counterpart in the earlier version, and is arguably the most defensible claim we can make, since it is the one we can demonstrate rather than assert.
- **Open.** Placement. This is long for an introduction bullet. Candidate: promote to a short standalone section before Sec. 3, so the methodology can be presented as answering D1–D4 explicitly.
- **Open.** Scope of the claim. Is the economics argument (hardware-test cost vs. simulation-test cost) something we can support in this paper, or a position statement we should flag as future work? It currently reads as the former.
- **Scope.** Single thruster plus rudder, under 7 m, field-identified parameters.
 - Underactuated rudder steering is the more interesting case: control authority is an explicit function of forward speed, so the steering and speed channels cannot be treated as independent design problems. Differential or holonomic thrust reframes the problem rather than simplifying it, and removes precisely this coupling.
 - It is also the more representative case for the installed base of small craft, even though the USV literature is dominated by differential-thrust research platforms.
 - Distinct as well from large vessels, where scale-model tow-tank testing is a reasonable route to parameters, so identification is not the binding constraint.
- **Contributions.**

- A three-level model spectrum methodology for this class of USV.
- Cross-model diagnostics that identify structurally salient dynamics.
- A margin-based sufficiency test (Diagnostic 4) that reports the operating point at which added model fidelity stops changing the control design, replacing a judgment call with a computed number.
- An empirical demonstration of when physical structure is required (black-box simulation failure under limited data).
- A targeted data collection framework motivated by diagnostic gaps.

2 Closely Related Work

Organizing axis: what is the least complex model structure that still supports control design, and what does the available excitation actually permit one to identify? Together these bound the cost of added fidelity, which is the quantity that is missing when complexity is chosen by fit alone.

2.1 Response Models as the Minimal Control-Oriented Description

- **Nomoto, Taguchi, Honda, Hirano (1957)** [9], *On the Steering Qualities of Ships*, Int. Shipbuilding Progress 4(35), 354–370. *No digital copy in circulation.*
 - **Point.** Origin of the first-order steering model $r(s)/\delta(s) = K/(Ts + 1)$. Collapses coupled sway-yaw into a gain and a time constant that map directly onto compensator decisions: T sets achievable closed-loop bandwidth, K sets loop gain. This is exactly our Level-1 structure, applied to surge as well as yaw.
 - **Open.** Developed for large displacement vessels. Does the first-order form hold for a sub-7 m hull? Sonnenburg’s data say yes at higher speeds; we should state this rather than assume it.
- **Fossen (2011)** [10][•], *Handbook of Marine Craft Hydrodynamics and Motion Control*, Wiley. doi:10.1002/9781119994138
 - **Point.** Canonical reduction of the linearized maneuvering equations to the Nomoto form and the source of our notation. Situates the response model in the broader hierarchy, which is the hierarchy we are traversing.
 - **Open.** We hold the 2011 first edition; a 2021 second edition exists (doi:10.1002/9781119575016). Cite whichever we actually work from.
- **Åström and Källström (1976)** [11][•], *Identification of Ship Steering Dynamics*, Automatica 12(1), 9–22. doi:10.1016/0005-1098(76)90064-9
 - **Point.** Found the identification side of this tradition: linear steering dynamics estimated from full-scale trial records by maximum likelihood. Supports our central claim that the binding constraint is experimental, not algorithmic.
 - **Open.** They use ML with a Kalman filter; we use ordinary least squares on a discrete ARX(1,1). Do we need to justify OLS, or is it self-evident at 10 Hz on a short clean record? Probably one sentence.
- **Källström and Åström (1981)** [12], *Experiences of System Identification Applied to Ship Steering*, Automatica 17(1), 187–198. doi:10.1016/0005-1098(81)90094-7 *No local copy. Try Lund LUP; the 1976 companion is free there.*
 - **Point.** Practical follow-on: the maneuvers a vessel can safely execute limit which param-

eters are distinguishable. This is the cleanest available statement of the excitation-limits-identifiability point.

- **Open.** Need the PDF to confirm this claim is actually made in these terms before we lean on it. Currently sourced from secondary description.

2.2 Physics-Based Maneuvering Models and the Identifiability Barrier

- **Abkowitz (1964)** [13], *Lectures on Ship Hydrodynamics: Steering and Manoeuvrability*, Report Hy-5, Lyngby. *No digital copy in circulation.*
 - **Point.** The canonical high-fidelity structure: third-order truncated Taylor expansion of hydrodynamic forces about steady straight-line motion. Cited for provenance as the pole opposite Nomoto.
 - **Open.** Cite directly, or via a secondary account (Fossen Ch. 7, Sonnenburg diss. Sec. 2.1) that we can actually read? Leaning secondary.
- **Yasukawa and Yoshimura (2015)** [14], *Introduction of MMG Standard Method for Ship Maneuvering Predictions*, J. Marine Sci. Tech. 20(1), 37–52. doi:10.1007/s00773-014-0293-y *Open access; Springer redirect blocked scripted retrieval.*
 - **Point.** The modular alternative to Abkowitz and the current standard for ship maneuvering prediction. Establishes that the high-fidelity pole is a live practice, not a historical artifact.
 - **Open.** Does MMG earn its space in a 6-page paper, or is Abkowitz alone enough to anchor the pole? Candidate cut if we run long.
- **Suárez, Berndt, Abdel-Maksoud (2026)** [1][•], *Regularized Machine Learning for System Identification of Ship Free-Running Manoeuvres from CFD-Based Synthetic Data*. arXiv:2606.17121
 - **Point.** *The citation our cost-of-fidelity argument rests on.* Abkowitz derivatives are strongly multicollinear; conditioning degrades as the coefficient set grows; the effective remedies are regularization and more diverse excitation, not better optimization. “Conditioning degrades as the coefficient set grows” is the cost of added fidelity stated as a measurable quantity.
 - **Point.** Their data are CFD-generated from idealized zigzag and turning maneuvers, which sharpens rather than weakens the point: if a coefficient set is ill-conditioned under noise-free synthetic excitation designed expressly for the task, it will not be recovered from a short trial on the water.
 - **Open.** Preprint, under review. If it has not landed by submission, restore Luo and Li (2017), doi:10.1016/j.apor.2017.06.008, alongside it rather than instead of it; that paper attributes parameter drift to dynamic cancellation among linear derivatives plus multicollinearity among nonlinear ones.
 - **Open.** Do we run Ridge on our own Level-2 parameters as a further diagnostic? Cheap to do, and it would make the citation reciprocal. Tracked at Sec. 3.2; note that Diagnostic 4 is already taken by the margin test.
- **Xu and Guedes Soares (2025)** [15][•], *Review of System Identification for Manoeuvring Modelling of Marine Surface Ships*, J. Marine Sci. Appl. 24, 459–478. doi:10.1007/s11804-025-00681-w
 - **Point.** Single citation covering “this difficulty is general, not specific to one method or one hull.” Saves us three or four individual citations.
 - **Open.** Confirm it addresses small USVs and not only ship-scale vessels. If ship-only, we must

say so rather than imply generality we have not checked.

- **Alexandersson, Mao, Ringsberg (2022)** [2][•], *System Identification of Vessel Manoeuvring Models*, Ocean Engineering 266, 112940. doi:10.1016/j.oceaneng.2022.112940
- **Point.** Identified coefficients need not be individually meaningful even when aggregate prediction is good. This separates “predicts well” from “parameters mean something,” which is precisely the interpretability axis our three levels are arranged along.
- **Open.** Confirm the vessel scale and data source (model tests vs. full scale) so we characterize the result accurately.

2.3 Control-Oriented Identification of Small Uncrewed Craft

- **Sonnenburg and Woolsey (2013)** [3][•], *Modeling, Identification, and Control of an Unmanned Surface Vehicle*, J. Field Robotics 30(3), 371–398. doi:10.1002/rob.21452
Companion: **Sonnenburg et al. (2010)** [16][•], OCEANS 2010 Seattle. doi:10.1109/OCEANS.2010.5664297 *Local copies are the 2013 dissertation (superset) and the extended VT report.*
- **Point.** *The nearest antecedent.* A 4.8 m planing RHIB; objective stated as a model “sufficiently rich for model-based control design, sufficiently simple to allow parameter identification.” Candidate speed and steering models identified and compared from field trials.
- **Point.** Two-stage identification: steady-state input-output maps from constant-input maneuvers first, then transient parameters with those gains fixed. Our two-phase separation of quadratic drag from thrust gain follows the same logic of estimating each parameter where it dominates. *Cite again in Sec. 3.2.*
- **Point.** Their conclusion is that no single model wins: Nomoto-with-sideslip below roughly 2.6 m/s, plain Nomoto above, deployed as a speed-scheduled combination.
- **Point.** *How they evaluate sufficiency* (answers the question raised in Sec. 1). Candidates are scored by a cost function evaluated on held-out validation maneuvers and tabulated per model and per regime, for both forward and sternward motion. Selection then applies a parsimony judgment: the full linear steering model wins marginally on cost in the stable regimes, but its two extra parameters are judged “not worth the slight edge in turn rate estimation.”
- **Point.** *This is the point we build on.* Their cost function measures the benefit of added parameters. Nothing in the procedure measures the cost, and no accepted metric for it exists, so the trade is necessarily settled by expert intuition. Read generously, this is the strongest prior work reaching the limit of the available apparatus rather than declining to use it. Our cross-model diagnostics are an attempt to put evidence on the vacant side of that scale.
- **Point.** Two differences define our contribution. Their vessel steers by vectored thrust, which removes the speed-dependent control effectiveness that dominates rudder-steered craft. And their comparison stops at structured models, with no data-driven level to bound residual complexity.
- **Open.** Their envelope reaches 22 kn with regime transitions; ours is 0–3 m/s in displacement only. Does their speed-scheduling conclusion transfer, or is it an artifact of crossing into planing?
- **Open.** Should we adopt their steady-state-first procedure for Level 1 too, rather than fitting ARX directly? Would make the two levels methodologically consistent.
- **Eriksen and Breivik (2017)** [17][•], *Modeling, Identification and Control of High-Speed ASVs:*

Theory and Experiments, LNCIS 474, 407–431. doi:10.1007/978-3-319-55372-6_19

- **Point.** Demonstrates the full path from field identification to deployed control: an 8.45 m ASV spanning displacement, semi-displacement, and planing, with four controllers tested in full scale. Evidence that control-oriented identification pays off, which is the premise we inherit.
- **Open.** At 8.45 m with rich instrumentation. Which parts of the method survive our sensor suite (GPS speed at 5 Hz, IMU yaw rate)? Worth one explicit sentence rather than silence.
- **Muske, Ashrafiuon, Haas, McCloskey, Flynn (2008)** [18][•], *Identification of a Control Oriented Nonlinear Dynamic USV Model*, ACC 2008, 562–567. doi:10.1109/ACC.2008.4586551
 - **Point.** *Direct precedent for our Level-2 method.* Drag parameters identified from dedicated deceleration and terminal-velocity tests, with thrust deliberately excluded from the regression. Same reasoning as our coasting-segment identification of d_2 . *Cite again in Sec. 3.2.*
 - **Open.** They use a model boat in controlled conditions. Does coast-down survive wind and current in the field? Our dataset has limited coasting; is one long deceleration enough, and how do we defend that?
- **Hann, Sirisena, Wongvanich (2010)** [19], *Simplified Modeling Approach to System Identification of Nonlinear Boat Dynamics*, ACC 2010, 5218–5223. doi:10.1109/ACC.2010.5530459 *No local copy.*
 - **Point.** Simplified identification of nonlinear boat dynamics with deliberately tractable experiments.
 - **Open.** Unclear how much this adds beyond Muske. Candidate cut if the overlap is large; decide once we have read it.
- **Wirtensohn, Reuter, Blach, Schuster, Hamburger (2013)** [20][•], *Modelling and Identification of a Twin Hull-Based Autonomous Surface Craft*, MMAR 2013, 121–126. doi:10.1109/MMAR.2013.6669892
 - **Point.** Concrete example of a complete identification campaign at small-ASV scale. Useful as the template we are deviating from.
 - **Open.** Differential thrust, so control effectiveness is not speed-dependent the way ours is. This supports our scope claim in Sec. 2.5; make sure we use it there rather than overstating its similarity here.
- **Morel, Orihuela, Bejarano (2023)** [21][•], *Modelling and Identification of an Autonomous Surface Vehicle*, MARTECH 2023, Castelló de la Plana.
 Journal version: **Morel, Orihuela, Combastel, Bejarano (2025)** [22][•], *Practical Identification Approach for the Actuation Dynamics of Autonomous Surface Vehicles with Minimal Instrumentation*, Ocean Engineering 319, 120098. doi:10.1016/j.oceaneng.2024.120098 / arXiv:2410.00631
 - **Point.** *Closest match to our instrumentation constraint.* Premise is that the sensor suite of a small autonomous craft, not the estimator, is the binding constraint. The 2023 workshop paper names the vessel (Yellowfish ASV) and tabulates parameters; the 2025 journal paper generalizes the method.
 - **Open.** They model *actuation* dynamics explicitly. Do we need an actuator model, or is our normalized RCOU-to-force mapping adequate? If we skip it, say why, because this paper makes the omission conspicuous.
- **Tanveer and Ahmad (2022)** [23][•], *Unmanned Surface Vehicle: Yaw Modeling and Identifica-*

tion, ICEANS 2022. [arXiv:2303.13064](#)

- **Point.** Honest illustration of what a single short experiment supports: a second-order yaw model from pool trials at 59% cross-validation fit. Supports our data-coverage claim from the low end.
- **Open.** Twin thruster, pool trials, two-page paper. Is it too weak to carry weight, or is its weakness exactly the point? Currently used as the latter.
- **Synthesis for this subsection.** The literature converges on simplified, field-identified models as the right instrument at this scale. What it does not do is *retain* more than one. Each study selects a structure, identifies it, and proceeds to control; comparison, where it occurs, is used to discard alternatives rather than to interrogate the dynamics.

2.4 Data-Driven and Grey-Box Alternatives

- **Woo, Park, Yu, Kim (2018)** [24], *Dynamic Model Identification of Unmanned Surface Vehicles Using Deep Learning Network*, Applied Ocean Research 78, 123–133. [doi:10.1016/j.apor.2018.06.011](#) No local copy. Highest-priority acquisition.
 - **Point.** LSTM reduces surge error 76.9% and yaw-rate error 60.7% versus linear models. Our Level-3 result directly challenges the generality of this headline, so the paper is the foil for our central empirical claim.
 - **Open.** *Must verify before we assert it:* do they report one-step-ahead prediction only, with no closed-loop simulation? Our Sec. 2.5 gap 2 depends entirely on this being true. Do not submit without checking.
- **Xu, Han, Cheng, Cheng, Ge (2022)** [25]^{*}, *A Physics-Informed Neural Network for the Prediction of Unmanned Surface Vehicle Dynamics*, JMSE 10(2), 148. [doi:10.3390/jmse10020148](#)
 - **Point.** Embedding 3-DOF structure in the training loss improves generalization from small datasets. Supports our “structure is regularization” claim from the opposite direction: they add structure to a network, we measure what its absence costs.
 - **Open.** How small is their “small”? If it is far larger than our 130 s, the comparison needs qualifying.
- **Zhang, Li, Xiong, He (2024)** [26]^{*}, *GBM-ILM: Grey-Box Modeling Based on Incremental Learning and Mechanism for Unmanned Surface Vehicles*, JMSE 12(4), 627. [doi:10.3390/jmse12040627](#)
 - **Point.** Represents the current hybrid state of the art: mechanism model plus incremental learning. Establishes that the field’s answer to data hunger is to reintroduce structure, which is our answer too, arrived at by a different route.
 - **Open.** Is this the best representative, or is there a stronger 2025–26 grey-box USV paper? One search before submission.
- **Xue, Wang, Qin, Deng (2022)** [27], *Probabilistic Identification of Unmanned Surface Vehicles Using Efficient Gaussian Processes with Uncertainty Propagation*, ICUS 2022, 1004–1009. [doi:10.1109/ICUS55513.2022.9986638](#) No local copy.
 - **Point.** An alternative answer to “how much should you trust this model”: propagate uncertainty so model confidence is available to the controller.
 - **Open.** Lowest-priority citation in the section. Cut if we run long.
- **Synthesis for this subsection.** Unstructured learning is data-hungry in exactly the way field

identification of a small USV cannot accommodate. Our Level-3 NARX is not a competitor to these methods but a deliberately unregularized reference point, whose closed-loop failure quantifies how much regularization the physical structure supplies.

2.5 Positioning of This Work

- **Gap 1: hierarchy as selection, not retention.** Prior work retains one level and discards the rest. We retain all three and use disagreement between them as a diagnostic that localizes which dynamics are structurally significant and which are simply unobserved. Against [3,17]^{••}.
 - **Open.** Is this alone strong enough for an OCEANS contribution, or does it read as methodology-only? Gaps 2 and 3 may be doing more work.
- **Gap 2: evaluation protocol.** Data-driven marine models are predominantly evaluated on one-step prediction, obscuring the closed-loop divergence that determines usefulness for control design. We report both metrics for the same model and dataset. Against [24].
 - **Open.** Contingent on the Woo verification above.
- **Gap 3: vessel class.** The small, single-thruster, rudder-steered vessel is underrepresented: closest prior work uses vectored thrust [3][•], differential thrust [20,23]^{••}, or a substantially larger platform [17][•]. Rudder steering makes control authority an explicit function of forward speed, the coupling our diagnostics target.
 - **Open.** Do we need a citation establishing that rudder-steered small USVs matter operationally, or is the platform’s existence sufficient warrant?
- **Framing.** The intended use is not only compensator synthesis but the simulation environments in which autonomy is developed and tested [28]. This aligns the work with identification for control [6–8], where model quality is judged against the design decision the model informs rather than fit statistics alone.
 - *Resolved:* the sim-to-real framing in Sec. 1 makes the simulator the primary consumer of these models, not a secondary one, so the citation is on-topic rather than incidental. Keep.

3 Model Spectrum Methodology

- **Framing.** Three levels, each identified from the same dataset. Each answers a different design question.

3.1 Level 1: First-Order Linear SISO Models

- Nomoto-style first-order transfer functions for surge and yaw, treated as independent SISO systems.
 - Surge: $G_v(s) = K/(\tau s + 1)$, input throttle (normalized), output speed.
 - Yaw: $G_r(s) = K_r/(\tau_r s + 1)$, input rudder, output yaw rate.
- **Identification.** Discrete-time ARX(1,1) by ordinary least squares; convert poles to physical τ and K .
- **Design utility.** Directly readable time constants and gains for PID tuning; reveals dominant timescale per channel.
- **Limitation.** Linear, decoupled; ignores quadratic drag, surge-yaw coupling, and operating-regime dependence. Alone, risks a compensator architecture mismatch.

- **Open.** Adopt Sonnenburg’s steady-state-first procedure here for consistency with Level 2? See Sec. 2.3.

3.2 Level 2: Nonlinear State-Space Model

- Physics-based two-state model from planar rigid-body equations:
 - Surge: $\dot{v} = k_T u - d_2 v |v|$.
 - Yaw: $\dot{r} = k_\delta \delta - d_r r$.
- Quadratic drag $v|v|$ is the key nonlinearity relative to Level 1.
- **Two-phase identification** (precedent: [3, 18]^{••}):
 - Phase 1 (coasting, $u \approx 0$): identify d_2 from zero-thrust deceleration segments.
 - Phase 2 (powered): identify k_T with d_2 fixed.
 - Decoupling avoids conflating thrust and drag in a joint fit.
- **Design utility.** Reveals the drag structure relevant to feedforward compensation and speed-hold steady state, while retaining physical interpretability.
- **Open.** Report a conditioning or regularization check on the joint fit, per [1][•], to make the identifiability argument concrete on our own data rather than only by citation.

3.3 Level 3: Black-Box NARX Model

- NARX multilayer perceptron predicting $[v(k+1), r(k+1)]$ from n lags of $[v, r, \text{throttle}, \text{rudder}]$. MIMO, so it can discover coupling the structured models assume away.
- **Architecture.** 3 lags, one hidden layer (20 neurons), ReLU, StandardScaler, early stopping.
- **Training.** Chronological 80/20 split (first 100 s train, last 30 s test) to avoid temporal leakage.
- **Evaluation.** One-step prediction *and* closed-loop simulation reported separately; these diverge significantly at this data scale.
- **Design utility.** Bounds the benefit of added complexity given available data; motivates richer collection; reference point as the dataset grows.

3.4 Cross-Model Diagnostics

- The spectrum’s value is the comparative analysis across levels, not the three fits individually.
- **Diagnostic 1: speed-dependent rudder gain.** Does replacing constant K_r with $K_r v$ improve the yaw-rate fit? Result: no improvement with current data ($\Delta R^2 = -0.03$), which flags a gap in data coverage rather than absence of the effect.
- **Diagnostic 2: drag nonlinearity.** Linear vs. quadratic fit to coasting deceleration. Result: quadratic dominant, $\Delta R^2 = +0.20$, $D_1 \approx 0$, $D_2 = 0.44 \text{ m}^{-1}$.
- **Diagnostic 3: rudder-induced surge drag.** Correlation of δ^2 with surge residuals. Result: negligible, $r = -0.08$; coupling term not warranted.
- **Diagnostic 4: is the added fidelity inside the robustness margin?** The first three diagnostics ask whether a term is *present* in the data. This one asks whether it *changes the design*, which is the question Sec. 1 argues has no accepted metric. It does have one: the robustness margin of the compensator is a budget for model error, so the test is whether the Level-1 to Level-2 discrepancy fits inside it.

- **Step 1.** Design a nominal compensator from Level 1 alone (PI on surge, P or PI on yaw). Record gain margin, phase margin and the complementary sensitivity $T(j\omega)$.
- **Step 2.** Linearize Level 2 about operating points spanning the tested envelope. For surge the quadratic drag gives a speed-scheduled first-order plant, $\tau(v_0) = 1/(2d_2v_0)$ and $K(v_0) = k_T/(2d_2v_0)$, so both the gain and the time constant fall as $1/v_0$. With the identified $d_2 = 0.438$ and $k_T = 3.963$ this is roughly $\tau : 1.1 \rightarrow 0.4$ s and $K : 4.5 \rightarrow 1.6$ across $v_0 = 1$ to 2.9 m/s, against Level 1’s fixed $\tau = 1.95$ s, $K = 6.42$.
- **Step 3.** Form the multiplicative error $\ell(\omega, v_0) = |G_2(j\omega, v_0)/G_1(j\omega) - 1|$ and test robust stability, $\|\ell T\|_\infty < 1$. The cheap version that will fit in a short paper is the gain ratio $K(v_0)/K_1$ against the gain margin.
- **Step 4, the useful output.** The test does not return yes or no. It returns *the speed at which Level 1 stops being sufficient*, which is an actionable number for a control designer and a direct answer to “enough for what.”
- **Expected findings, to be confirmed.** For yaw, Level 2 reduces to $r/\delta = 0.771/(0.523s+1)$ against Level 1’s $0.771/(0.471s+1)$: the same DC gain and an 11% difference in time constant, so ℓ is small and the test should pass across the envelope. Level 2 buys nothing for yaw control, and we should say so. For surge the gain falls by roughly a factor of four across the envelope, so the test should fail at the upper end. Note the failure mode is sluggishness rather than instability, because quadratic drag is stabilizing: the Level-1 design is conservative at speed, not unsafe.
- **Open.** Scope. The full $\|\ell T\|_\infty$ treatment may not fit this paper. Minimum viable version: report $K(v_0)/K_1$ against the Level-1 gain margin and name the crossover speed. Full treatment flagged as future work.
- **Implication.** Diagnostics 1–3 inform the Level-2 structure and specify what the next data collection must target. Diagnostic 4 converts the sufficiency question from a judgment into an arithmetic one, and is the piece that lets this work claim a method rather than a position.

4 Experimental Platform and Data Collection

- **Vehicle.** [model/name], approx. [length] m, single propeller plus rudder, ArduPilot autopilot, GPS and IMU.
 - Inputs: RC throttle (RCOU_C3) and rudder (RCOU_C1), 1100–1900 μ s PWM.
 - Outputs: GPS speed (5 Hz), IMU yaw rate (137 Hz).
- **Data cleaning.** Multi-rate streams aligned to a 10 Hz common grid by linear interpolation; RC commands normalized to $[-1, 1]$; one 29.3 s gap (vessel repositioning) handled by segmentation.
- **Current dataset.** Single step-response trial, 130 s working window (150–280 s of a 341 s recording), throttle 0–39%, speed 0–2.9 m/s. Intentionally minimal: a proof of concept for the workflow.
- **Planned dataset (full paper).** Multiple trials spanning throttle 0–100%, dedicated rudder sweeps at fixed speeds to resolve speed-dependent gain, and maneuvers spanning displacement, semi-displacement, and planing. [describe vehicle speed range and transition speeds]

5 Results

5.1 Level 1: First-Order SISO

- Table: identified τ and K for surge and yaw.
- Simulation R^2 for both channels.
- Figure: measured vs. simulated speed and yaw rate.
- Note: parameters directly usable for PID design.

5.2 Level 2: Nonlinear State-Space

- Table: identified k_T , d_2 , k_δ , d_r .
- Compare d_2 from coasting vs. joint fit; motivate why two-phase is necessary.
- Simulation R^2 comparison vs. Level 1.
- Figure: measured vs. simulated, combined plot across all three levels.

5.3 Level 3: NARX

- One-step prediction R^2 vs. closed-loop simulation R^2 ; highlight the gap.
- Figure: simulation divergence for surge.
- Interpretation: limited data makes black-box models unreliable simulators; physical structure is essential regularization at this data scale.

5.4 Cross-Model Diagnostic Summary

- Summary table: four diagnostics, findings, implication for model and control design.
- Diagnostic 4 reported in its minimum viable form: $K(v_0)/K_1$ against the Level-1 gain margin, plus the crossover speed at which Level 1 stops being sufficient for surge.
- Key message: the spectrum reveals structure that no single model exposes, and the margin test says which of that structure changes the design.

6 Discussion

- **What the spectrum reveals for this vehicle class.**
 - Surge and yaw are weakly coupled at the speeds tested, so decoupled SISO controllers are a reasonable starting point.
 - Quadratic drag is significant: speed-hold controllers need nonlinear feedforward; linear controllers will show speed-dependent steady-state error.
 - Speed-dependent rudder gain can neither be confirmed nor denied. This is the primary gap the expanded dataset must fill. If confirmed, gain scheduling or adaptive control is warranted.
- **Implications for compensator architecture.**
 - Level 1 $\rightarrow$ baseline PID structure with identified gains.
 - Level 2 $\rightarrow$ nonlinear feedforward for drag on the surge channel, where the margin test is expected to show Level 1 failing at speed. For yaw the two levels nearly coincide, so Level 2 changes nothing and we should say so.

- Gain scheduling on the rudder loop remains open, gated on Diagnostic 1 rather than on Level 2.
- Level 3 → motivates MPC or learning-based control once data suffices.
- **Fidelity vs. insight: return to D1–D5 (Sec. 1).**
 - State plainly that none of the three levels reaches D5. Even the highest-fidelity model here cannot eliminate experimental fine-tuning, and we should say so rather than let the reader infer otherwise.
 - The defensible claim is D1 through D4: the spectrum tells you which architecture to build, lets you verify it end to end, supports relative comparison between candidates and tells you where in the parameter space field tuning will actually be needed so that the tuning workflow can be built in advance.
 - **Open.** Is “the model does not remove field tuning, but it tells you what to tune and lets you rehearse it” a strong enough payoff to headline? It is honest, and it is the question most modeling papers decline to answer.
- **Limitations** of the current dataset and what the full study adds.

7 Conclusion

- Summarize the three-level spectrum methodology for small underactuated USVs.
- Restate the core finding: the spectrum collectively provides more control design guidance than any single model; simple models are not a stepping stone but a persistent design tool.
- State what the full dataset will allow: resolution of speed-dependent rudder dynamics, validation across operating regimes and a more definitive comparison of black-box vs. structured models.
- **Future work: margin-based sufficiency.** Diagnostic 4 is reported here in its minimum viable form. The extension is to treat the robustness margin as the formal sufficiency criterion across the whole spectrum, computing for each design decision the model level at which further fidelity is provably inert. That would replace the expert judgment identified in Sec. 2.3 with a computable bound, and it generalizes past this vehicle class.
- Release the dataset and analysis notebooks as open tools for the USV research community.

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